

CSS Phase 8: Advanced CSS

Introduction

Advanced CSS-এর মূল উদ্দেশ্য:

- ✓ Clean Code লেখা
 - ✓ Reusable CSS তৈরি করা
 - ✓ Maintainable Project বানানো
 - ✓ Modern CSS Techniques শেখা
-

1. CSS Variables

Variables ব্যবহার করে একই Value বারবার না লিখে Reuse করা যায়।

Creating Variables

```
``css id="p7m2k8":root{
```

```
--primary-color:#2563eb;  
--secondary-color:#10b981;  
--text-color:#333;
```

```
}
```

```
---  
  
## Using Variables  
  
``css id="m4q8v1"  
h1{
```

```
color:var(--primary-color);  
}
```

Button Example

```
```css id="n8k3p6" button{
```

```
background:
var(--primary-color);

color:white;
```

```
} ```
```

---

## Benefits

### Without Variables

```
css id="j6m2x9"
color:#2563eb;
```

```
background:#2563eb;
```

```
border-color:#2563eb;
```

একই Color বারবার লিখতে হবে।

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### With Variables

```
```css id="f3p8k5" var(--primary-color)
```

একবার Change করলেই সব জায়গায় Update হবে।

2. calc()

CSS-এর ভিতরে Calculation করার জন্য।

Example:

```
```css id="w5m9q2"
width:calc(100% - 40px);
```

Meaning:

```text id="r8k1p7" 100% থেকে 40px বাদ

Another Example:

```
```css id="s2m7v4"
height:calc(100vh - 80px);
```

---

## 3. min()

ছোট Value নির্বাচন করে।

```
```css id="k9p4m8" width:min(500px, 90%);
```

Browser ছোট Value ব্যবহার করবে।

4. max()

বড় Value নির্বাচন করে।

```
```css id="t4q8x2"
width:max(300px, 50%);
```

---

## 5. clamp()

Responsive Typography-এর জন্য খুব জনপ্রিয়।

Structure:

```
```css id="h6m1p9" clamp( minimum, preferred, maximum )
```

Example:

```
```css id="q3k8m5"
font-size:
clamp(
1rem,
5vw,
3rem
);
```

---

Benefits

Desktop:

```
```text id="u8p2x4" বড় Font
```

Mobile:

```
```text id="z4m7q1"
ছোট Font
```

Automatically Responsive ।

---

## 6. Object Fit

Image Control করার জন্য ।

---

**Cover**

```
```css id="v1k6m8" object-fit:cover;
```

Container পূর্ণ করবে।

Contain

```
```css id="f8p3q5"
object-fit:contain;
```

Image পুরো দেখা যাবে।

---

Example:

```
```css id="m2k7v9" img{
```

```
width:100%;
height:300px;
object-fit:cover;
```

```
} ```
```

7. Aspect Ratio

Image বা Box-এর Ratio Fix রাখে। `css id="x5m8p1"`
`aspect-ratio:16/9;`

Example:

```
```css id="a4q7k2" .video{
```

```
aspect-ratio:16/9;
```

```
} ```
```

---

## 8. Scroll Behavior

Smooth Scroll তৈরি করে। `css id="p9m3v7"`

```
html{
 scroll-behavior:smooth;
}
```

---

Example:

```
```html id="k2x8m5" <a href="#about"> About </a>
```

Click করলে Smooth Scroll হবে।

9. Scroll Snap

Modern Slider-এর জন্য।

Container:

```
```css id="q8m4p1"  
.container{
 scroll-snap-type:x mandatory;
}
```

---

Items:

```
```css id="v7k2m9" .item{
```

```
    scroll-snap-align:start;
```

```
}```
```

10. Advanced Selectors

:not()

নির্দিষ্ট Element বাদ দিতে। `css id="m5p8q3"`

```
div:not(.active){  
  
    opacity:.5;  
}
```

:is()

একাধিক Selector একসাথে।

```
``css id="a9m2k7" :is(h1,h2,h3){
```

```
    color:blue;
```

```
} ``
```

:where()

:is() এর মতো কিন্তু Priority কম। `css id="f4q8p2"`

```
:where(h1,h2,h3){
```

```
    margin-bottom:10px;
```

```
}
```

11. CSS Nesting

Modern CSS Feature

আগে:

```
``css id="x2m7k9" .card h2{
```

```
    color:red;
```

```
} ``
```

এখন: `css id="p5q8m1"`

```
.card{
```

```
  h2{
```

```
    color:red;
```

```
  }
```

```
}
```

12. Container Queries

Component Responsive করার Modern Technique ।

```
``css id="v8m3q6" @container (max-width:400px){
```

```
  .card{
```

```
    flex-direction:column;
```

```
  }
```

```
}
```

```
---
```

Media Query পুরো Screen দেখে।

Container Query শুধু Container দেখে।

```
---
```

13. Logical Properties

Responsive ও Multi-language Website-এর জন্য।

পুরনো:

```
``css id="m1k8q5"
```

```
margin-left:20px;
```

Modern:


```
```css id="q6m3p9" margin-inline-start:20px;
```

```

```

## # 14. CSS Layers

Large Project-এর জন্য।

```
```css id="k3q8m2"
@layer base,
components,
utilities;
```

CSS Organize করতে সাহায্য করে।

Modern Responsive Card Example

```
```css id="z8m5p4" .card{
```

```
width:min(100%, 400px);

padding:20px;

border-radius:12px;
```

```
}
```

```
.card img{
```

```
width:100%;

aspect-ratio:16/9;

object-fit:cover;
```

```
}```
```

---

# Most Important Advanced CSS Topics

css id="w4m9q8"

CSS Variables

var()

calc()

min()

max()

clamp()

object-fit

aspect-ratio

scroll-behavior

:not()

:is()

:where()

container queries

---

## Phase 8 Practice Projects

### Beginner

1. Modern Button System
2. Responsive Cards
3. Smooth Scroll Website

### Intermediate

1. Portfolio Website
2. Blog Layout
3. Product Showcase

## Advanced

1. SaaS Landing Page
  2. Admin Dashboard
  3. Complete Responsive Website
- 

# Phase 8 Checklist

☐ CSS Variables

☐ var()

☐ calc()

☐ min()

☐ max()

☐ clamp()

☐ object-fit

☐ aspect-ratio

☐ smooth scroll

☐ scroll snap

☐ :not()

☐ :is()

☐ :where()

☐ CSS Nesting

☐ Container Queries

☐ Logical Properties

☐ CSS Layers

☐ Responsive Components

Phase 8 শেষ করলে তুমি Modern CSS-এর বেশিরভাগ গুরুত্বপূর্ণ Feature ব্যবহার করতে পারবে এবং Professional Level Frontend Development-এর জন্য প্রস্তুত হবে।